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STUDYBROS

PROBLEM

- Students have the material (slides, PDFs, recordings) but not the time to turn it into usable study assets
- Recall alone isn't mastery; most tools don't track whether it was understood or just saw it once
- Existing study apps require a separate account, a separate app, and a separate habit

REQUIREMENTS

- Students converting lecture PDFs/slides into study material without manual prep
- Self-learners generating quizzes from downloaded or recorded material
- Core: ingest PDF/PPTX/TXT/MD/video, generate flashcards/quizzes/tests, schedule reviews by actual retention, run under 30 seconds per generation call

ARCHITECTURE

AI STUDY GUIDE "StudyBros"		
FRONTEND	Web UI (Next.js): Upload · Documents · Generate · Study · Quiz/Test player · Progress · Achievements CLI (Click): ingest, generate, export, list, status MCP Client: any MCP-compatible LLM (Claude, ChatGPT, Cursor)	
API	FastAPI REST (api/routes/): documents.py upload, list, rename, delete, bulk-delete, chunks generate.py flashcards, quiz, practice-test, summary, providers study.py review, due, schedule, session, mastery, progress study_sets.py list, detail, delete gamification.py phase, achievements, strengths-weaknesses, consistency export.py export by study set MCP Server (mcp_server/server.py): 10 tools · 9 resources · 8 prompts — mirrors the REST surface for MCP clients, generation delegated to the connected LLM (no key needed)	
APPLICATION	Ingestion: FileScanner, Extractors (PPTX/PDF/TXT/MD/Video), TextChunker Generation: Schemas (Pydantic v2), Prompts, Generator (OpenAI/Anthropic/OpenRouter)	Learning: FSRS Scheduler (scheduler.py) Gamification engine (gamification.py) Export: JSONExporter, AnkiExporter, MarkdownExporter
DATABASE	sources · documents · chunks · study_sets card_reviews · study_sessions · user_progress · achievements · gamification_state	
EXTERNAL	OpenAI GPT / Anthropic Claude / OpenRouter — generation OpenAI Whisper — audio/video transcription FFmpeg — audio extraction	

OUR SOLUTION

- Ingest → chunk → AI-generate flashcards, quizzes, and practice tests from a student's own material
- FSRS v5 spaced repetition + Bloom's-taxonomy mastery tracking
- Delivered three ways from one core library: web app, MCP server (use it from Claude/ChatGPT/Cursor with zero API key), and CLI Provider-agnostic generation (OpenAI, Anthropic, OpenRouter) — never locked to one vendor

RESULTS

- 211 automated backend tests passing, clean lint and type-checking across the codebase
- Live-verified end-to-end: upload → generate → review → export, including real 20MB+ file uploads
- Two structured UI/UX audits completed (WCAG 2.1 AA + performance + security), 152 findings identified and triaged; evaluation was real, not a formality
- Shipped a fully working platform after losing a team member mid-project